M.E. CSE PROJECT REVIEW · MODULES 1 AND 2

Romanised Tamil queries break RAG retrieval and transliteration repairs most of it

Code-mixed RAG: chunking, embedding and query normalisation for Tamil–English (Tanglish) queries

STUDENT

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GUIDE

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REVIEW

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Module 1

Tanglish retrieval for RAG is an open gap

Problem, literature, objectives and design

Tamil users type Tanglish but RAG retrievers are built and tested in English.

"scholarship ku last date enna?" No prior study evaluates RAG retrieval components for Tamil–English code-mixed and romanised queries.

One question arrives in four forms that share almost no characters

FORM	EXAMPLE QUERY
English	Who is responsible for the Israel-Hamas conflict
Tamil script	இஸ்ரேல் ஹமாஸ் மோதலுக்கு யார் பொறுப்பு
Romanised Tamil	Israel Hamas mothalukku yaar poruppu
Code-mixed	Israel Hamas conflict ku yaar responsible

Prior work covers mixed-script Hindi or native Tamil but never Tanglish RAG

WORK	LANGUAGES	WHAT IS MISSING
FIRE MSIR 2013–16; Gupta et al. SIGIR 2014	Hindi, Bengali retrieval; Tamil in labelling only	No Tamil retrieval; pre-dense
Chandu et al. 2018 (W18-3204)	Tamlish, Hinglish, Tenglish	Factoid QA, not RAG
FIRE CMIR 2024–25	Bengali–English	Bengali social media
Zeng et al. 2026	Chinese, Japanese + English	Not Tamil or romanised
Balasubramaniyam 2026	Sinhala, Tamil → English	Native script only
nlpctx dataset 2026	Tamil, 4 query forms	No paper or baselines

Four objectives take the study from baseline to published benchmark

01

Build a four-form query set over Tamil documents

02

Measure the drop and test transliteration fixes (done in Module 2)

03

Compare a Tamil-aware chunker on longer documents (Module 3)

04

Measure generated-answer quality and publish (Modules 4–5)

The proposed pipeline adds the script handling existing RAG lacks

Existing RAG

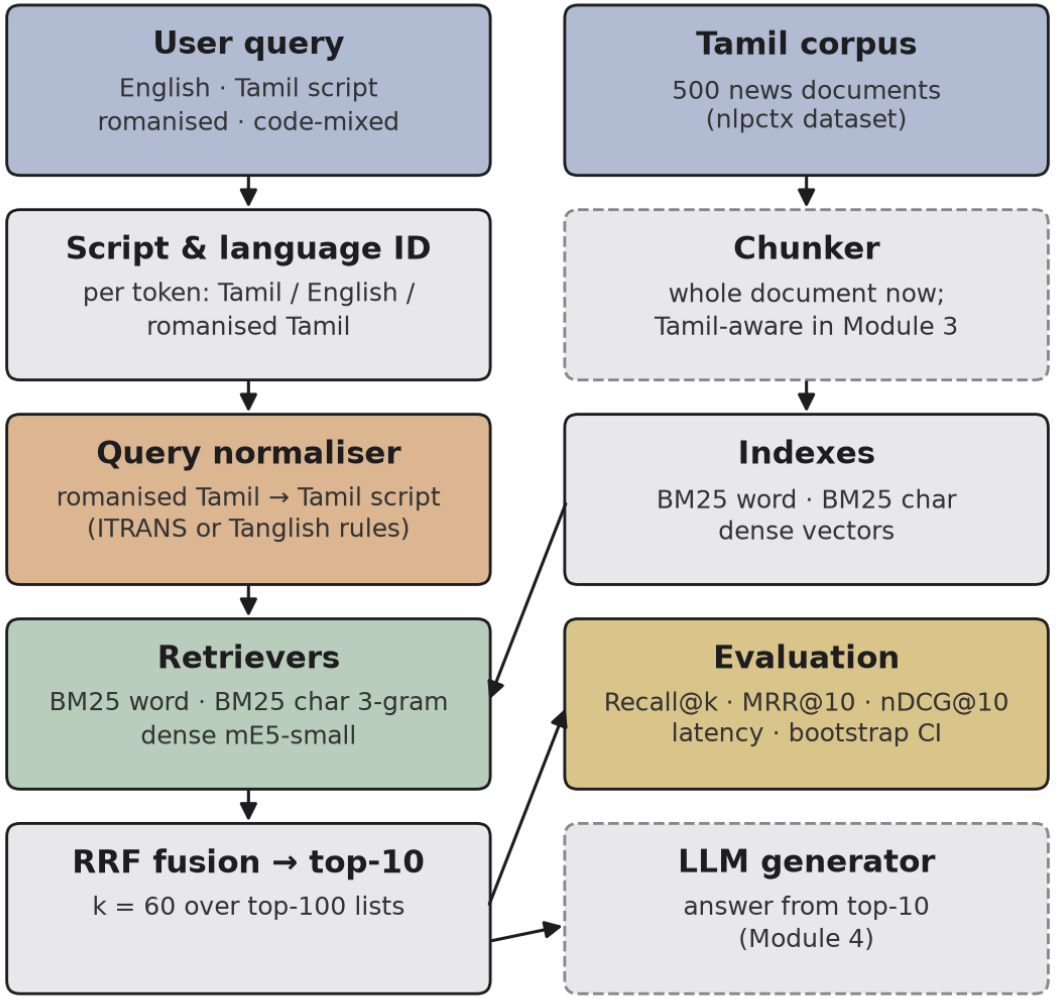
Query used as typed. Word BM25 plus dense. RRF fusion by default. Tested in English only.

Proposed

Token language ID. Romanised Tamil transliterated to Tamil script. Char 3-gram BM25. Fusion measured per query form.

The query is normalised before three retrievers and rank fusion

Solid boxes are built in Module 2. Dashed boxes come in Modules 3 and 4.



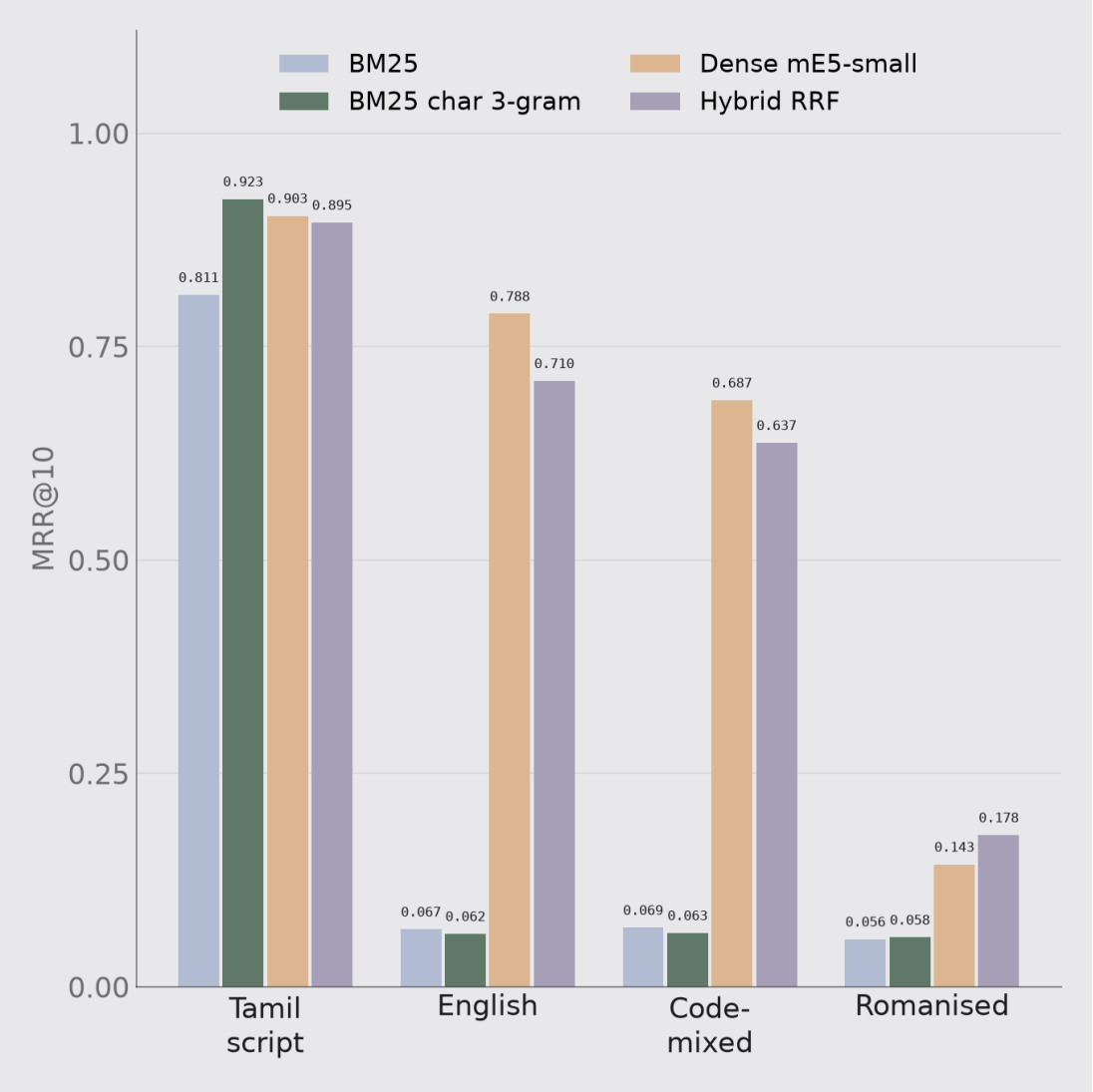
Module 2

Sixty real retrieval runs on a 2-core CPU

500 Tamil news documents · 2,154 queries per form · 8 query sets · 60 main runs + 24 naive-fusion reruns

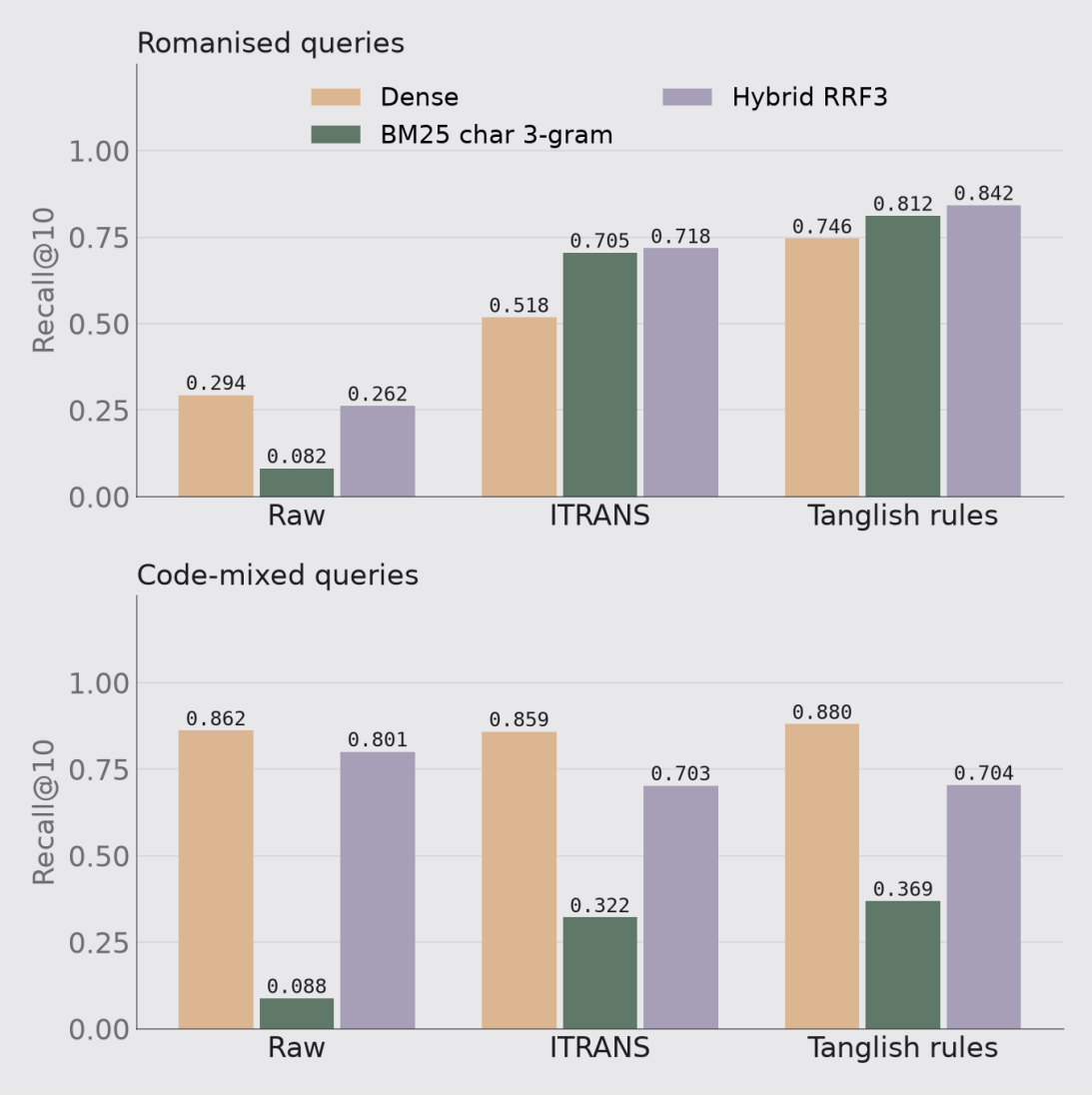
Romanised queries fall below MRR 0.18 while Tamil script reaches 0.92

Romanised Tamil: dense Recall@10 is only 0.294 and the best raw MRR 0.178. Seven in ten questions miss.



Tanglish rules lift romanised Recall@10 from 0.08 to 0.81

Rules + char 3-gram BM25: MRR 0.058 → 0.684 ($p < 0.001$, also under cluster bootstrap) for 0.3 ms per query.

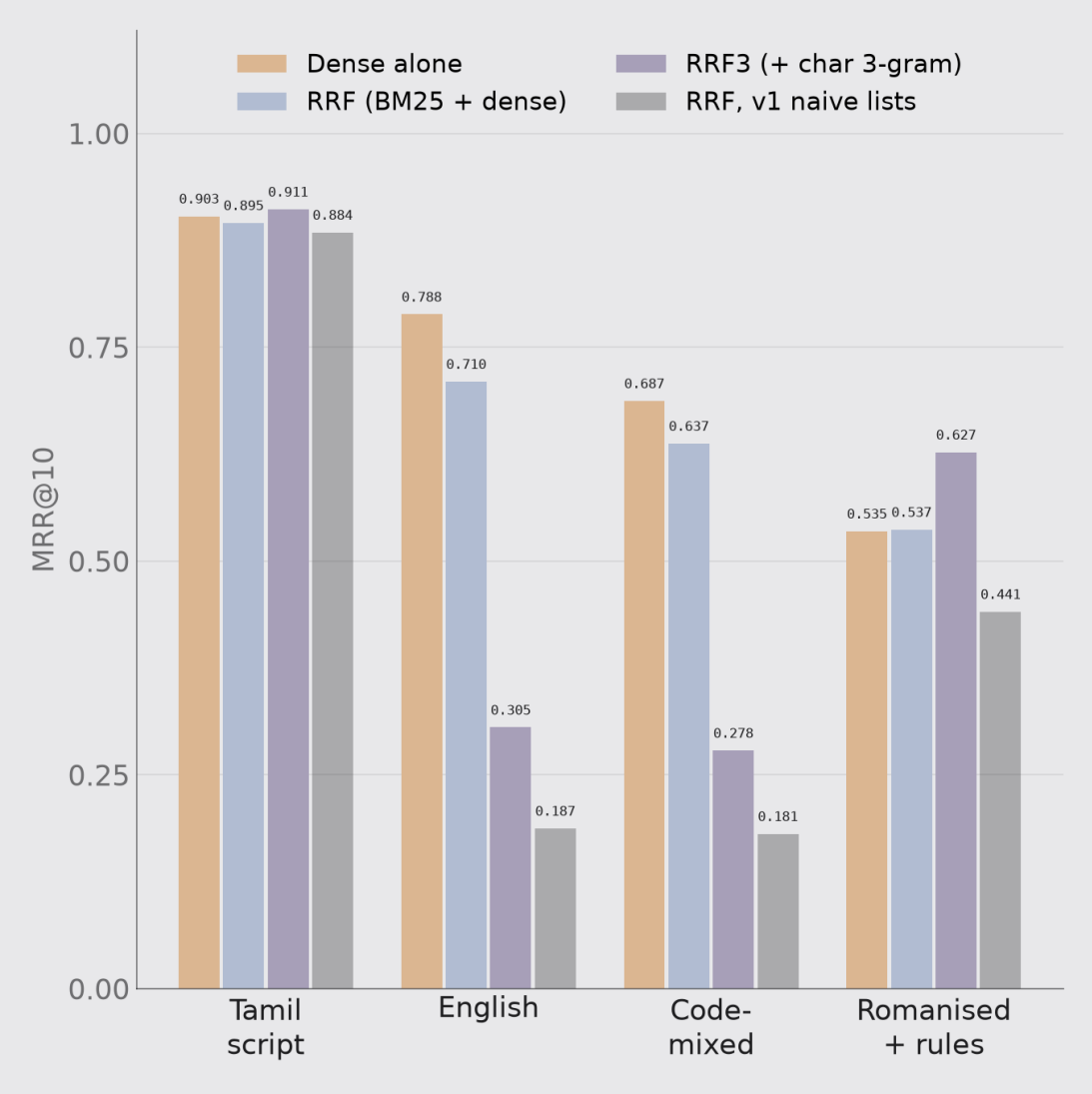


Transliteration rescues real queries but misses unmarked long vowels

QUERY GIVEN TO THE RETRIEVER	RANK OF GOLD
thakkudhalil Colombia raanuva thiraigal uyirizhappu	not in top 10
தக்குதலில் Colombia ரானுவ திரைகல் உயிரிழப்பு (rules)	1
Vadakku maagaana paadasaalai seerudai neelam korikkai	not in top 10
வடக்கு மாகான பாடசாலை சீருடை நீலம் கொரிக்கை (rules)	1
Muthalaq thadai masoda Rajyasabha	not in top 10
முதலக் தடை மசொட ரஜ்யசப (rules)	not in top 10

Equal-weight fusion still cuts English-query MRR from 0.79 to 0.71

Corrected after a bug: zero-score BM25 documents had entered RRF and overstated the loss as 0.79 → 0.19. Gate fusion by script.



The dataset's code-mixed queries are mostly English

22%

No Tamil word at all

13%

Identical to English

1.7

Tamil tokens per query

NEXT

Module 3 builds a script-gated router and tests it on 500 human Tanglish questions

Then Module 4 adds LLM answer evaluation and Module 5 the paper and public release.

DEC–JAN

Human queries, G.O.s, IndicXlit and LLM baselines

FEB–MAR

Generation-side RAG evaluation

APR–MAY

Paper, thesis and release